Hugging Face Setup Guide

Hugging Face Account · API Token · Google Colab Secrets · Gemma Models

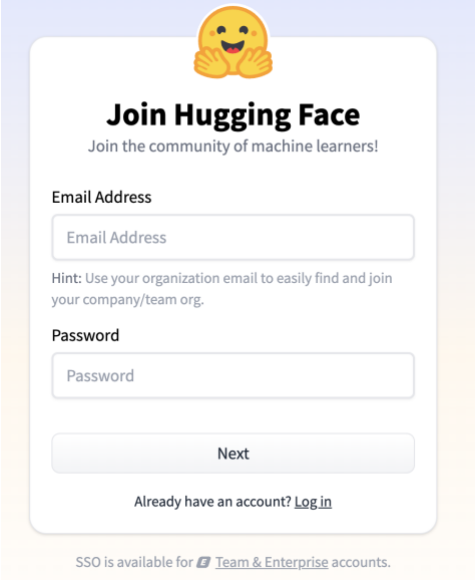
Overview

This guide provides step-by-step instructions for configuring the tools required to download and run Google Gemma language models within Google Colab notebooks. Complete all five steps in order before using Google Colab for the practical session.

#	Step	Notes
1	Create a Hugging Face account	*huggingface.co/join — free, one-time*
2	Generate an access token (HF_TOKEN)	*Settings → Access Tokens → Read role*
3	Accept the Gemma model licence	*Required for all gated Gemma variants*
4	Add HF_TOKEN to Colab Secrets	*🔑 Secrets panel — Notebook access ON*
5	Authenticate from the notebook	*Two lines of Python at the top of the notebook*

1 Create a Hugging Face Account

huggingface.co/join — free registration



Navigate to huggingface.co/join follow and complete the registration form.

a. In successive steps, enter a **Email address**, **Password, Username**.

b. Open the verification email sent to your address and confirm it by clicking the link provided.

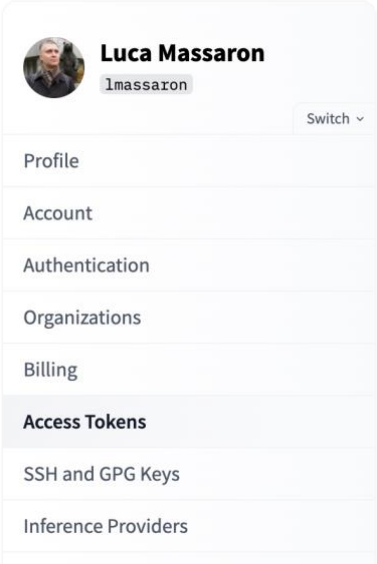
NOTE

Use a permanent email address. Your credentials will be required in subsequent practical sessions.

2 Generate an Access Token (HF_TOKEN)

Settings → Access Tokens — Read permission

The access token is the credential used to authenticate API requests and model downloads programmatically.



a. After logging in, click your **profile picture** (top-right corner) → **Settings**.

b. In the left-hand navigation, select **Access Tokens**.

c. Click **+ Create new token**. Assign a recognisable name (e.g. `colab-token`) and set the role to **Read** .

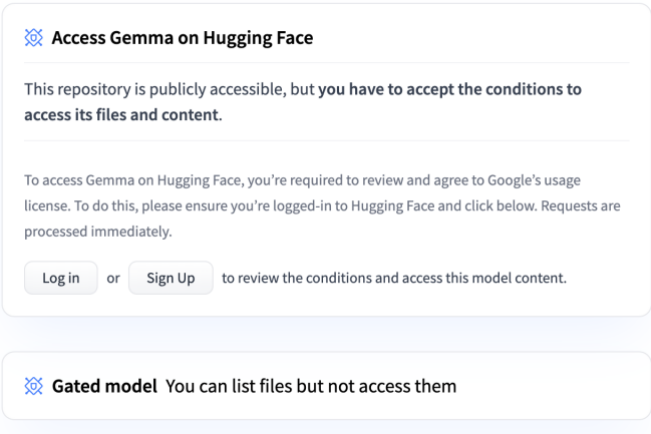
d. Click **Generate a token**. A string beginning with `hf_` will be displayed. **Copy it immediately** — it will not be shown again.

IMPORTANT — Token Security

Treat the token as a password. Do not embed it as plain text in a notebook, commit it to a repository, or share it with others.

3 Accept the Gemma Model Licence

One-time per model variant — mandatory for gated models



Google Gemma models are **gated**: access requires explicit acceptance of the terms of use on the model page. This is the most frequent cause of authentication errors when downloading the model.

a. Navigate to the model page for the variant specified by the notebook you want to use, e.g.: huggingface.co/google/gemma-2b.

b. A notification banner will state that you must agree to share your contact information to access the model.

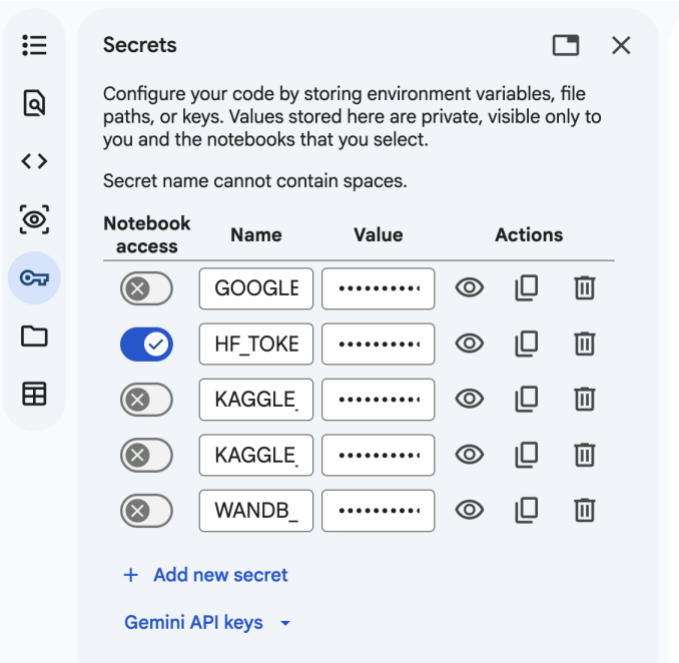
c. Complete the short form if prompted, then click **Acknowledge licence**.

d. Confirm that a green **"You have access to this model"** banner appears before proceeding.

> **NOTE**
> If the course uses multiple Gemma variants (e.g. `gemma-2b` and `gemma-7b-it`), the licence must be accepted on each model page separately.

4 Add HF_TOKEN to Google Colab Secrets

Secrets panel — Notebook access must be enabled



Colab Secrets provide a secure mechanism to store credentials without exposing them in notebook cells. This approach is strongly preferred over hardcoding the token in code.

a. Open a notebook at colab.research.google.com.

b. In the left sidebar, click the 🔑 **key icon** to open the Secrets panel.

c. Click **+ Add new secret**.

d. Set the **Name** field to exactly `HF_TOKEN` (uppercase, no spaces).

e. Paste your token (`hf_…`) into the **Value** field.

f. Enable the **Notebook access** toggle. The slider must be active (blue) before closing the panel.

> **REMINDER**
> The Notebook access toggle resets for new notebooks. Verify it is enabled each time you open a different notebook that requires the token.

5 Authenticate from the Notebook

Add the following cells at the top of your notebook

To test if everything works or for your own project, place the following code in the first cell of any notebook that requires access to Hugging Face models.

5.1 Authenticate with Hugging Face

```
# Read the token from Colab Secrets and log in to Hugging Face
from google.colab import userdata
from huggingface_hub import login

hf_token = userdata.get('HF_TOKEN')
login(token=hf_token)
```

5.2 Load a Gemma Model

Once authenticated, load the model as shown below. Replace the model identifier as instructed:

```
from transformers import AutoTokenizer, AutoModelForCausalLM

model_name = "google/gemma-2b"   # adjust to the required variant

tokenizer = AutoTokenizer.from_pretrained(model_name)
model     = AutoModelForCausalLM.from_pretrained(model_name)
```

TROUBLESHOOTING — 401 Unauthorised Error

If a 401 error occurs during model download, verify the following:

1. The Gemma model licence has been accepted (Step 3).
2. The Notebook access toggle is enabled in Colab Secrets (Step 4f).
3. The Secret name is spelled exactly as `HF_TOKEN` (case-sensitive).

Completion Checklist

Verify each item before the practical session:

- Hugging Face account created and email address verified.
- Access token generated with Read permission and securely stored.
- Gemma model licence accepted on the Hugging Face model page.
- `HF_TOKEN` added to Colab Secrets with Notebook access enabled.
- Notebook cell runs